Affine and Projective Geometry

Twenty chapters have studied the vector space, and a vector space has an origin. Geometry does not. Nothing a geometer says — these three points are collinear, these two lines are parallel, this point is the midpoint of that segment — mentions a zero vector, and the sets that geometry actually hands us are usually not subspaces at all. Chapter 3's solution set of $\mathbf{Ax} = \mathbf{b}$ with $\mathbf{b} \neq \mathbf{0}$ is the first example in the book: it misses $\mathbf{0}$, the sum of two of its points is not one of its points, and yet it is a perfectly good flat piece of space. Something is being described there that the word "subspace" does not reach.

There are two classical repairs, and this chapter is both of them. **Affine geometry** deletes the origin and keeps what survives: points, differences of points, flats, parallelism, ratios along a line, and the maps that send lines to lines. **Projective geometry** goes the other way and *adds* points — one for every direction — so that two lines in a plane always meet, the dimension formula for a join loses its second case, every quadric in a plane is one of a very short list, and interchanging "point" with "hyperplane" turns every theorem into another theorem.

Everything here is proved with linear algebra, and mostly with the linear algebra of Chapters 1 to 5. An affine space is a vector space acting freely and transitively on a set; a projective space is the set of lines through the origin of a vector space; a projectivity is an invertible linear map read on those lines; a conic is a quadratic form; a line in space is a wedge of two vectors. The geometry is the payoff, not the method.

22.-1.1 Conventions

- **Points are not bold.** P, Q, R are points of an affine space $\mathcal{A}$, and $\overrightarrow{PQ} \in V$ is the vector from P to Q; vectors follow the book's rule by type, bold in coordinates, $\mathbf{x} \in F^n$, and plain in an abstract space, $v \in V$. We write $\mathbb{A}^n$ for affine n-space, the set F^n with its origin deliberately forgotten.
- **A flat is an affine subspace**, and $\vec{A}$ is its direction space, a subspace of V. The join $A \vee B$ is the smallest flat containing both, the affine counterpart of $U + W$.
- $\mathbb{P}(V)$ **is the projective space of** V, of dimension $\dim V - 1$, with $\mathbb{P}^n(F) = \mathbb{P}(F^{n+1})$; $[\mathbf{x}] = [x_0 : x_1 : \cdots : x_n]$ is the point determined by $\mathbf{x} \neq \mathbf{0}$. The empty projective space has dimension -1, and that convention is used, not merely tolerated. The projective linear group is $\mathrm{PGL}(V)$.
- **The cross-ratio** $(A, B; C, D)$ **is a point of** $\mathbb{P}^1(F)$, not a scalar. Over a field where the affine chart applies it is the familiar number, and Section 7 gives the formula; keeping it projective is what lets it take the value $[1 : 0]$ without apology.
- V^* **is Chapter 5's dual space** and $\mathbb{P}(V^*)$ the dual projective space, whose points are the hyperplanes of $\mathbb{P}(V)$. Section 8 writes S^0 for the subspace of $\mathbb{P}(V^*)$ dual to S, built from the annihilator.
- $Z(q)$ **is the zero set of a quadratic form**, and "conic" means $Z(q)$ in a projective plane. Section 5's affine quadric is instead the zero set of a degree-two polynomial $\mathbf{x}^\top \mathbf{A}\mathbf{x} + 2\mathbf{b}^\top \mathbf{x} + c$, and carries the $(n+1) \times (n+1)$ matrix $\widetilde{\mathbf{A}}$ alongside the small matrix $\mathbf{A}$.

22.-1.1.1 Which field, and where the characteristic bites

Sections 1, 2, 6, 7, 8 and 9 work over an **arbitrary field** F, with no hypothesis on the characteristic at all. That includes the chapter's finite-field computations: the four-point affine plane over $\mathbb{F}_2$ in Section 1, and the count of $\mathbb{P}^n(\mathbb{F}_q)$ with the Fano plane in Section 6.

Characteristic 2 **is barred exactly where quadratic forms appear**, which is Section 5

throughout and Section 10 up to the Plücker material. The reason is Chapter 14's: over $\mathbb{F}_2$ a quadratic form is not determined by a symmetric matrix, and Section 5 opens by showing that there is then nothing left to classify — every $\mathbf{x}^\top\mathbf{A}\mathbf{x}$ with $\mathbf{A}$ symmetric collapses to the square of a linear form. Section 5 names the hypothesis at each of the three places it does work, and Section 10 states it at the head and lifts it explicitly for the Plücker subsection, which needs no assumption on F. Two smaller restrictions are local and flagged where they occur: Section 2's centroid needs $3 \neq 0$ as well as $2 \neq 0$, and over $\mathbb{F}_3$ the three medians really are parallel; Section 7's negative result about $n+1$ points needs F to have at least three elements, and Section 7 proves that over $\mathbb{F}_2$ the negative result is false.

Three sections are metric, and they are over $\mathbb{R}$ with the standard inner product: Section 4 entirely, the second half of Section 5, and Section 3, which inherits Chapter 18's setting of $\mathbb{R}^n$ with the dot product. Everywhere else a distance would be meaningless: a projectivity does not preserve one.

22.-1.2 What the chapter quotes, and what it does not prove

The chapter imports no analysis. It develops none of its own — no limits, no continuity, no compactness argument — and it adds nothing to the list of analytic facts **(A1)**–**(A6)** that Chapter 16 introduced. It uses that list exactly once, at one remove: Section 3's proof that a bounded polyhedron is the hull of its vertices cites Chapter 16's Corollary 16.2.5 to supply the compactness hypothesis of Chapter 18's Theorem 18.8.5. That is the whole of it. Section 3 says so at its head, and uses the words "closed", "bounded", and "compact" only inside the results it cites.

Topology is likewise absent. $\mathbb{P}^2(\mathbb{R})$ is a famous topological object, and a reader who has met it will expect the words "compact" and "nonorientable"; Section 6 says plainly that this chapter does not use them, because every argument here is a statement about subspaces of a vector space, valid over any field, including finite ones where topology has nothing to say.

Six things are quoted and not proved. Each is flagged again where it appears, and **nothing in the chapter rests on any of them.**

- **Section 9** — the coordinatization theory of axiomatic projective planes: that Desargues's theorem holds in a plane exactly when the plane comes from a division ring, that Pappus's theorem holds exactly when that ring is commutative, that Pappus therefore implies Desargues, and that non-Desarguesian planes exist. This book has built neither division rings nor synthetic planes, and "projective plane" has meant $\mathbb{P}(V)$ throughout. Section 9 states these with credit in one remark and proves Desargues and Pappus itself, over an arbitrary field, by linear algebra.
- **Section 8** — the duality principle is claimed as a **working method, not a meta-theorem**. What Section 8 proves is a list of five equivalences about the correspondence $S \mapsto S^0$; to turn "every incidence statement has a dual" into a theorem one has to define a formal language and induct on it, which this book has not set up. So every dual statement the chapter uses is proved individually, and Section 8 says this in as many words.
- **Section 4** — the classification of motions is proved in full for $\mathbb{R}^2$ and $\mathbb{R}^3$, both the direct and the opposite cases. The general shape in $\mathbb{R}^n$ — a translation along a flat together with mutually orthogonal rotation planes — is described in a warning and is **not** proved; the warning says so, and gives $\mathbf{R}_{\pi/2} \oplus \mathbf{R}_{\pi/3}$ in $\mathbb{R}^4$ as the witness that the three-dimensional list does not extend.
- **Section 3** — that a dual variable is the *exact* rate at which the optimal value gains when

a constraint is loosened. Section 3 proves the inequality, that y_i **bounds** the gain, and then exhibits a two-row program where the exact reading fails. It also declines to develop the finite description of an unbounded polyhedron as a polytope plus a cone, pointing at Chapter 18's exercise instead.
- **Section 10** — *which* quadric the Klein quadric is. The main text records that it is smooth, with inertia $(3, 3, 0)$ over $\mathbb{R}$; the Gram matrix that shows this is computed in Exercise (C2: The Klein quadric), and nothing in the section rests on the answer.
- **Section 10** — which proportionality factors turn the parametrization of the circle into *primitive* Pythagorean triples. That is arithmetic about parities and common factors, not linear algebra, and Section 10 says so where it stops.

Two exercises from earlier chapters are cited, both **for credit only**: Chapter 15 §10's Exercise (C1: The Plücker relation), which reaches the Plücker relation by the $\omega \wedge \omega$ route, and Chapter 18 §09's Exercise (C1: Unbounded polyhedra). Section 10 proves the Plücker relation in both directions from results in the text, and its proof needs no hypothesis on the characteristic, which the exercise's route does. No theorem here rests on an exercise.

One debt is paid. Chapter 15 §08 hinted at Plücker coordinates without naming the chapter that would build them; Section 10 does.

22.-1.3 What you need

The chapter leans hard on Chapters 0 to 5 and on Chapters 12, 14, and 18, and lightly on everything else. **Chapters 9, 10, 13, 19, 20, and 21 are not used at all.** Eigenvalues appear only through Chapter 12's spectral theorem and Chapter 14's inertia; no canonical form of an operator, no positive semidefinite matrix, no singular value, and no matrix inequality is needed anywhere.

- **Chapter 0** — sets, maps, and the algebra the constructions rest on: functions and their inverses for Section 4, equivalence relations and quotient sets for Section 6's construction of $\mathbb{P}(V)$, the characteristic of a field for Section 5, and the group axioms, which the affine, Euclidean, and projective groups each verify: Theorem 0.4.7, Theorem 0.4.2, Definition 0.5.2, Definition 0.5.5, Theorem 0.5.2, Definition 0.7.2, Theorem 0.9.3, Theorem 0.9.2, Definition 0.10.1, Definition 0.10.2, Definition 0.10.3, Example (Groups), Theorem 0.10.1, Theorem 0.10.3.
- **Chapter 1** — the chapter's constant companion, and above all the dimension formula for a sum of subspaces, which is what both the affine join formula and the projective dimension formula reduce to: Theorem 1.2.1, Theorem 1.2.2, Theorem 1.3.1, Definition 1.4.1, Lemma 1.5.3, Theorem 1.5.1, Theorem 1.6.5, Theorem 1.6.8, Theorem 1.6.4, Theorem 1.6.3, Theorem 1.7.1, Theorem 1.7.4, Theorem 1.7.3, Theorem 1.8.2.
- **Chapter 3** — the object that opens Section 1, the solution set of a system with $\mathbf{b} \neq \mathbf{0}$; the rank facts Sections 2, 3 and 10 count with; and the observation that a homogeneous system with more unknowns than equations has a nontrivial solution, which is the whole existence half of "five points determine a conic": Theorem 3.1.1, Theorem 3.1.3, Corollary 3.2.5, Theorem 3.4.4, Theorem 3.5.7, Theorem 3.5.8, Theorem 3.5.5, Lemma 3.7.2.
- **Chapter 2** — linear maps, which are the linear parts of affine maps and the representatives of projectivities: Example (Translation), Theorem 2.1.6, Theorem 2.1.5, Theorem 2.2.1, Theorem 2.2.2, Theorem 2.2.3, Theorem 2.2.5, Theorem 2.2.4, Theorem 2.3.1, Corollary 2.3.2, Theorem 2.4.2, Theorem 2.5.1, Theorem 2.5.7, Theorem 2.5.3, Theorem 2.5.4, Corollary 2.5.5.
- **Chapter 4** — the rank of a map read off its matrix, and the cosets that flats turn out to be: Theorem 4.1.5, Definition 4.4.2, Lemma 4.4.3, Theorem 4.4.5, Theorem 4.5.1.

- **Chapter 5** — the whole of Section 8 and half of Sections 2 and 6. The annihilator is what makes projective duality a bijection, and the hyperplane-as-kernel-of-a-functional is what makes a flat a solution set: Definition 5.1.2, Theorem 5.1.3, Theorem 5.1.1, Proposition 5.1.2, Corollary 5.1.4, Definition 5.2.1, Theorem 5.2.3, Theorem 5.2.2, Corollary 5.3.4, Corollary 5.3.5, Theorem 5.3.2, Definition 5.4.1, Theorem 5.4.1, Definition 5.5.2, Theorem 5.5.3, Theorem 5.6.2, Lemma 5.6.1, Corollary 5.6.6, Proposition 5.6.4, Example (Equations for a plane in $\mathbb{R}^4$).
- **Chapter 6 §01** — one fact: a nonzero polynomial of degree at most d over a field has at most d roots, which is how an exercise of Section 5 knows that a line meeting a quadric in three points lies inside it. Section 10 uses Chapter 3 §07's version of the same bound for the same purpose, and that one is in the text: Corollary 6.1.4.
- **Chapter 7** — determinants, for the direct-or-opposite dichotomy of Section 4 and for the coordinate computations; the Vandermonde determinant is how Section 7 certifies a list of points in general position: Corollary 7.3.3, Theorem 7.4.5, Theorem 7.5.4, Theorem 7.5.2, Theorem 7.8.1, Corollary 7.8.2, Theorem 7.10.3.
- **Chapter 8 §01** — block multiplication, which is what the $(n+1) \times (n+1)$ matrix of an affine map is for: Theorem 8.1.1.
- **Chapter 11** — the Euclidean sections' toolkit: the inner product and Cauchy–Schwarz, orthogonal decomposition, the characterizations of an isometry, and the polarization identity that turns distance-preservation into linearity: Theorem 11.1.5, Theorem 11.1.2, Theorem 11.2.1, Corollary 11.2.3, Theorem 11.3.2, Theorem 11.6.5, Lemma 11.7.2, Theorem 11.7.3, Proposition 11.7.4, Theorem 11.7.6, Definition 11.8.1, Proposition 11.8.2.
- **Chapter 12** — the real spectral theorem, which rotates a quadric onto its principal axes, and the canonical form of an orthogonal matrix, which is the entire engine of Section 4's two classification theorems: Corollary 12.5.2, Theorem 12.5.5, Example (Finding the axes of a conic), Theorem 12.6.1, Corollary 12.6.2, Theorem 12.6.4, Theorem 12.6.5, Theorem 12.6.7.
- **Chapter 14** — the classification of quadratic forms, which Sections 5 and 10 turn into the classification of quadrics. Congruence is what an affine change of coordinates does to the big matrix, and Sylvester's law is what separates an ellipse from an empty set: Definition 14.1.2, Proposition 14.1.2, Definition 14.2.1, Proposition 14.2.2, Proposition 14.2.4, Theorem 14.2.3, Definition 14.3.2, Theorem 14.3.5, Theorem 14.3.3, Proposition 14.3.2, Theorem 14.4.2, Proposition 14.4.3, Corollary 14.4.4, Theorem 14.5.1, Corollary 14.5.2, Definition 14.5.1, Theorem 14.5.3, Definition 14.7.1, Corollary 14.9.3, Example (The Pfaffian in Sizes 2 and 4).
- **Chapter 15 §§08–10** — exterior powers, for Plücker coordinates alone: the basis of $\Lambda^2 F^4$, that a wedge is nonzero exactly when the factors are independent, and that its coordinates are the minors: Lemma 15.8.1, Theorem 15.8.5, Theorem 15.8.7, Corollary 15.8.6, Theorem 15.9.5, Exercise (C1: The Plücker relation) (credited, not used).
- **Chapter 16** — two facts, both in the metric sections: the reverse triangle inequality for Section 4's nonexample, and the compactness of closed bounded sets, which is the chapter's only contact with analysis: Lemma 16.1.1, Corollary 16.2.5.
- **Chapter 17** — the Rayleigh quotient, which reads Section 5's semi-axes off the eigenvalues, and the inertia triple: Proposition 17.1.1, Definition 17.10.1.
- **Chapter 18** — all of Section 3, and the affine vocabulary of Sections 1, 2 and 5. Convex hulls and affine hulls, separation, extreme points and faces, polytopes and Minkowski–Weyl, and linear programming duality: Definition 18.1.5, Proposition 18.1.4, Theorem 18.1.3, Proposition 18.1.1, Definition 18.1.2, Definition 18.2.1, Lemma 18.2.1, Theorem 18.3.6, Definition 18.6.2, Theorem 18.6.1, Theorem 18.6.3, Corollary 18.6.2, Theorem 18.6.5, Definition 18.8.1, Definition 18.8.2, Proposition 18.8.4, Lemma 18.8.3, Theorem 18.8.5, Corollary 18.8.6, Definition 18.9.2, Proposition 18.9.1, Theorem 18.9.5, Theorem 18.9.2, Corollary 18.9.3, Example (The cube and the cross-polytope), Exercise

(C1: Unbounded polyhedra) (pointed at, not used).

22.-1.4 Roadmap

- **Affine Spaces and Affine Maps.** A set on which a vector space acts freely and transitively: differences of points are vectors, sums of points are meaningless, and Chasles's relation is the one law. Choosing an origin identifies the space with V, and the identification is canonical only after the choice. Affine combinations are exactly the combinations whose coefficients sum to 1, and no other coefficient sum survives a change of origin. An affine map is a linear map plus a translation, for a unique linear part, and the whole affine group fits in one $(n+1) \times (n+1)$ block matrix — the device the rest of the chapter computes with.
- **Flats, Independence, and Coordinates.** Flats are the cosets, parallelism is an equivalence relation on flats of equal dimension, and two flats either meet or lie in two distinct parallel hyperplanes. The dimension of a join has two cases: the subspace formula when they meet, and a corrected formula with a $+1$ when they miss, in which the intersection of **directions** takes over. No convention for $\dim \emptyset$ merges the two. Then affine frames, barycentric coordinates, and an affine map pinned down by its values on an affine frame of $n+1$ points; on any set lying inside a proper flat — in particular on any n points — two distinct affine maps agree. The medians of a triangle meet at the centroid, in one line of barycentric coordinates — and over $\mathbb{F}_3$ they do not meet at all.
- **Convex Polyhedra, Geometrically.** A reading of Chapter 18, not a reproof: it says so in its first paragraph. A face is what a supporting hyperplane cuts out; faces are polyhedra and there are finitely many; vertex, extreme point, and basic feasible solution are three names for one thing. The face lattices of a cube and of a simplex are computed in full. A linear program is a family of parallel hyperplanes sweeping across a solid, the optimal set is a whole face, and the dual variables are the nonnegative weights expressing the objective as a combination of the active outer normals. Negative cases: an unbounded slab with no vertex at all, whose optimum is nonetheless attained on a face that contains none, and a quadrant that has a vertex and no optimum. No algorithm appears here or anywhere in this book; Chapter 24 is where algorithms live.
- **Euclidean Motions.** Distance alone forces everything: a map of $\mathbb{R}^n$ preserving distances is affine with orthogonal linear part, and **surjectivity is not assumed** — it is proved. Polarization is the move, and the Euclidean norm is doing real work, as a distance-preserving nonaffine map for the maximum norm shows. Motions form a group with the translations normal in it and the orthogonal group as the quotient's content, recorded without quotient groups, which this book has not built. Then both classifications in full: in the plane, translation, rotation, reflection, glide reflection; in space, translation, rotation, screw, reflection, glide, rotatory reflection, told apart by the fixed set. From $\mathbb{R}^4$ on the list breaks, and the section says so.
- **Affine and Euclidean Quadrics.** A quadric is a quadratic form that has forgotten where the origin is. An affine substitution acts on the $(n+1) \times (n+1)$ matrix by congruence, a center exists exactly when the linear vector lies in the column space of the small matrix, and the parabola is precisely the centerless case. Over $\mathbb{R}$ every quadric is affinely equivalent to exactly one of three families, separated by the pair of inertias; for $n = 2$ that is nine normal forms, two of which have empty zero sets and are still inequivalent, and for $n = 3$ it is seventeen. Allowing only motions instead keeps the eigenvalues, and they are the reciprocal squared semi-axes — the Rayleigh quotient read on an ellipsoid.
- **Projective Space.** Stop working with the points of a vector space and start working with its lines through the origin. The affine chart splits $\mathbb{P}^n(F)$ into a copy of $\mathbb{A}^n$ and a hyperplane at infinity of directions, and the two parallel lines of the opening now meet, at a point whose

coordinates are computed. The dimension formula $\dim(S \vee T) + \dim(S \cap T) = \dim S + \dim T$ has **no exceptional case**, which is the whole reason for the construction; two distinct lines of a projective plane meet in exactly one point. Finite fields give $|\mathbb{P}^n(\mathbb{F}_q)| = (q^{n+1} - 1)/(q - 1)$, and the Fano plane is drawn and checked.

- **Projective Transformations.** An invertible linear map descends to $\mathbb{P}(V)$, and two of them give the same projectivity exactly when they differ by a nonzero scalar — so $\mathrm{PGL}(V)$ is $\mathrm{GL}(V)$ modulo scalars, said without quotient groups. The fundamental theorem: $n + 2$ points in general position can be carried to any other such list by exactly one projectivity. $n + 1$ points are not enough, and the explicit failure is a diagonal matrix; over $\mathbb{F}_2$, and only there, $n + 1$ points do suffice, which is proved. Then the cross-ratio, invariant under projectivities and a complete invariant of four points on a line, computed exactly; perspectivities; and the sentence that ties the chapter together — affine maps are exactly the projectivities fixing the hyperplane at infinity as a set.
- **Duality.** A point of $\mathbb{P}(V^*)$ is a hyperplane of $\mathbb{P}(V)$. The annihilator gives an inclusion-reversing bijection $S \mapsto S^0$ between the projective subspaces of the two spaces, with $\dim S^0 = n - 1 - \dim S$, exchanging joins with meets and the two extreme subspaces with each other. Dualizing twice returns the original. The duality principle is then stated as five equivalences and used as a recipe, each application being its own proof; "two distinct points lie on a unique line" dualizes to "two distinct lines meet in a unique point", and the projective dimension formula turns out to be self-dual. The trap is spelled out: the dual of "point" is "hyperplane", and in $\mathbb{P}^3$ two lines need not meet.
- **Desargues and Pappus.** Two triangles are perspective from a point if and only if they are perspective from a line, proved inside $\mathbb{P}^2$ by choosing representatives so that the hypothesis reads $a' = a + o$, with every scaling step verified — that is where such proofs go wrong. The converse costs no second theorem: the same lemma, applied to a pair of auxiliary triangles built out of the same picture. Pappus follows in coordinates, with the frame chosen by the previous section's fundamental theorem to make the computation short, and the degenerate configurations excluded by hypothesis. Both hold over every field, and the remark that closes the section records, with credit and without proof, that Desargues means "division ring" and Pappus means "commutative".
- **Conics and Lines in Space.** Over an algebraically closed field a projective quadric is classified by rank alone; over $\mathbb{R}$ by rank and inertia up to sign, giving five conics in $\mathbb{P}^2(\mathbb{R})$: empty, a point, a line, two lines, a smooth conic. Five points in general position lie on exactly one conic, and the independence of the five conditions is proved, not assumed. A smooth conic with one point is in bijection with $\mathbb{P}^1$ by projection from that point, which turns one Pythagorean triple into all of them. Pole, polar, and the dual conic use Section 8's duality. Finally Plücker coordinates, over any field: a line of $\mathbb{P}^3$ is a point of $\mathbb{P}^5$, the six coordinates satisfy $p_{01}p_{23} - p_{02}p_{13} + p_{03}p_{12} = 0$, and **every** point of that Klein quadric is a line — both directions proved. A summary closes the chapter.

22.-1.5 Named moves

- **Choose an origin, then say that you chose it.** An affine space becomes a vector space the moment a point is picked, and every affine statement is then a linear statement about the difference vectors. The discipline is to record which conclusions survive a different choice: those are the affine ones, and the coefficient sum 1 is exactly the test.
- **Add a coordinate to linearize.** An affine map of F^n is not linear, but $\mathbf{x} \mapsto (1, \mathbf{x})$ — Section 1 writes the extra coordinate last instead — makes it the restriction of a linear map of F^{n+1}, with matrix $\left(\begin{smallmatrix} 1 & \mathbf{0}^\top \\ \mathbf{b} & \mathbf{M} \end{smallmatrix}\right)$. The same extra coordinate is what turns a quadric polynomial into a quadratic form and an affine space into a chart of a projective space. Three of this chapter's theorems are that one observation in three costumes.

- **Pass from points to lines through the origin.** Scaling is the enemy of a well-defined point and the friend of a well-defined condition: $q(\lambda x) = \lambda^2 q(x)$ vanishes when $q(x)$ does. Anything homogeneous descends to $\mathbb{P}(V)$, which is why every object in the second half of the chapter is the solution set of homogeneous polynomial equations.
- **Choose the representatives so the hypothesis becomes an equation.** A projective point is a whole line of vectors, and the content of most projective proofs is the scaling chosen at the start: $a' = a + o$ in Desargues, a normalized frame in the fundamental theorem, the second intersection point in the conic parametrization. State out loud which scalings remain free afterwards.
- **Split the translation and ask what $\mathbf{I}-\mathbf{Q}$ can absorb.** To classify a map $\mathbf{x} \mapsto \mathbf{Q}\mathbf{x}+\mathbf{b}$, decompose $\mathbf{b}$ along the kernel of $\mathbf{I} - \mathbf{Q}$ and its complement. The part in the image is absorbed by moving the fixed set; the part in the kernel survives and is the glide. That single question produces both of Section 4's classification theorems.
- **Read an old theorem geometrically instead of proving a new one.** Section 3 adds no convexity theorem to Chapter 18. It renames the objects — face, vertex, edge, facet — and the theorems become statements about a solid with flat sides. Knowing when a chapter is a translation rather than a construction is worth as much as the construction.
- **Count the conditions, then prove they are independent.** Five points impose five linear conditions on a 6-dimensional space of forms, so a conic exists by counting alone. Uniqueness is the real theorem, and it is where general position earns its keep: a second independent solution would let us push a conic through a sixth point on one of the lines, a conic meeting a line three times contains it, so it splits into two lines, and the second line then carries three of the original five. The count is cheap; the independence is the content.
- **Dualize by the dimension formula, never by the words.** The dual of a d-dimensional subspace has dimension $n-1-d$. Compute that number before translating any statement: in a plane it sends points to lines, but in $\mathbb{P}^3$ it sends points to planes and lines to lines, and translating "point" to "line" there produces a false theorem.